# Histogram Matching

Continuous example

Image histogram

$$P_r(r) = \begin{cases} \frac{3r^2}{(L-1)^3}, & 0 < r < L-1 \\ 0, & \text{otherwise} \end{cases}$$

$$P_z(z) = \begin{cases} \frac{2}{L-1} - \frac{2z}{(L-1)^2}, & 0 \le z \le L-1 \\ 0, & \text{otherwise} \end{cases}$$

$$S = (L-1)\int_0^r \frac{3w^2}{(L-1)^3}\,dw = \frac{r^3}{(L-1)^2}$$

$$G(z) = (L-1)\int_0^z P_z(t)\,dt = 2z - \frac{z^2}{L-1} = s \rightarrow \text{transformation} = \text{target}$$

$$\boxed{z = f(s)} \rightarrow \text{Goal}$$

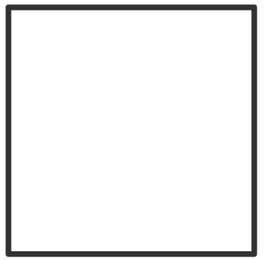
$$\frac{z^2}{L-1} - 2z + s = 0 \quad \times L-1 \quad \Rightarrow z^2 - (L-1)\,2z + s(L-1) = 0$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = (L-1) \pm \sqrt{(L-1)^2 - s(L-1)}$$

must only be $\ominus$ to fall within $0 < r < L-1$

Mapping function $\rightarrow$ $$z = (L-1) - \sqrt{(L-1)^2 - \frac{r^3}{(L-1)^2}(L-1)}$$

Discrete example using application $\rightarrow$



1. Select Hist specification
2. choose i/p & target randomly & take SS
3. Run & take SS
4. Calculate & send full document

prerequisite $\rightarrow$ pages 145, 146 ex 3.7 in DIP 4th

## Home Task

to get $G(z) = \text{CDF}.(\underset{\text{range}}{L-1})$